LESSON PLAN

ROCKET ENGINE (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Selection criteria for matrix material
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Selection criteria for matrix material. Different types of matrix material and brief description of Thermosetting matrix material and Thermoplastic matrix material.
5. Specific Objectives	17.1.1.1 Trainee will be able to understand the development of composite materials, types and functions of fibers and resin, Defects in composite, selection criteria for Matrix, fabric weaves, prepress, strength flowcharts

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of Matrix Material Selection Flowchart; Instructor Supervision: Verify trainee understanding of Thermoset vs Thermoplastic properties.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the matrix in a composite material?
(b) Name two common types of matrix materials used in composites.
(c) Why is the selection of matrix material critical for structural integrity?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Development of Composite Materials • Functions of Fibers and Resin • Common Defects in Composites • Selection Criteria for Matrix Material	L & Ds	PPT, IB	15 min
(b)	• Thermosetting Matrix Material Description • Thermoplastic Matrix Material Description • Comparison of Thermoset and Thermoplastic Properties • Impact of Matrix on Composite Performance	L & Ds	PPT, PDF	20 min
(c)	• Fabric Weaves and Prepress Techniques • Strength Flowcharts for Matrix Selection • Application of Selection Criteria in Design • Review of Defects Related to Matrix Choice	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the key difference between the curing process of thermosetting and thermoplastic matrices.
(b) How does the selection of a specific matrix material influence the defect profile of the final composite?

12. Home Assignment and Follow Up	Prepare a brief note on how the knowledge of matrix selection criteria will assist in the study of Prepregs in the next lesson plan.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Definition of Prepregs. Application of prepregs in aerospace industry. Advantages and disadvantages of prepress. Storage of Prepregs.	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min